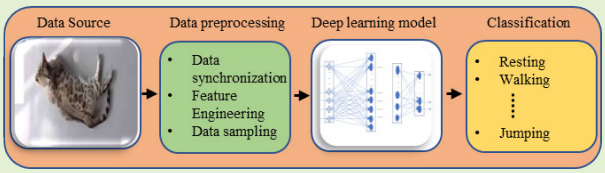
A Deep Learning Approach for Detecting and Classifying Cat Activity to Monitor and Improve Cat's Well-Being Using Accelerometer, Gyroscope, and Magnetometer

Ali Hussain, Sikandar Ali, Moon-Il Joo, and Hee-Cheol Kim

***Abstract*—Activity detection of pets using sensor technology and artificial intelligence (AI) approaches has gained momentum in recent years. These technologies have revamped up the monitoring systems for the well-being of pets and are emerging as revolutionary technologies. Most of the research works have been conducted on the activity detection of humans, dogs, sheep, and horses; however, cat activity detection has not been explored meticulously using the aforementioned latest technologies. In this article, we proposed an long short-term memory (LSTM)-based deep learning model and investigated the activity detection of cats using three wearable sensors. Our research content can be summarized as follows. First, we collected 1 765 825 data samples from ten cats of different ages, breeds, and gender using an accelerometer, gyroscope, and magnetometer. Second, the raw data were preprocessed by applying data transformation approaches and feature engineering techniques. Third, we developed three different kinds of deep learning models, namely; artificial neural network (ANN), 1-D convolutional neural network (CNN), and LSTM, to classify ten different activities of cats. Hyperparameter tuning was performed to optimize the performance of the models. LSTM model outperformed other models and showed good accuracy. The model achieved an accuracy of 95.94%. To the best of our knowledge, this work is the first of its kind, which applied a deep learning-based model for the detection of cat activity using three kinds of sensor data, such as accelerometer, gyroscope, and magnetometer. This proposed model will help improve the well-being of cats by detecting the activities of cats with high accuracy using wearable sensor devices and deep learning technology.**



***Index Terms*—Accelerometer, cat activity detection, deep learning, gyroscope, long short-term memory (LSTM), magnetometer, pet activity detection, sensor data, wearable sensors.**

I. Introduction

RECENTLY, human and pet activity recognition have drawn much attention for monitoring and tracking of their activities, which helps to understand behavioral changes, thus improving their overall health and well-being [1], [2], [3], [4]. The behavioral changes and activity patterns of humans and animals can be monitored using digital information processing technologies, which include wearable sensor technology, video recording devices, and artificial intelligence (AI)-based models [5], [6], [7], [8], [9], [10], [11]. These technologies help us to monitor and track the physical movement and body postures of animals with high precision. In the past, the activities and behaviors of pets were used to be monitored by human observers, which was not a good approach because it interfered with the normal life of pets; thus, the actual behavior could not be observed or captured.

Nowadays, the wearable sensor technology is being used for collecting activity data from pets. Wearable sensor technology enables continuous sensing of the daily activities of pets without interrupting their normal daily life behavior. These wearable sensors are attached to the body of the pet, and data are continuously generated and collected without any external

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This work involved human subjects or animals in its research. The authors confirm that all human/animal subject research procedures and protocols are exempt from review board approval.

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interference. Activity data collected through wearable sensor devices play a vital role in the development of AI-based pet activity models with optimal performance because there is a minimal chance of activity data being generated by any external interference. Thus, this will lead to predicting the actual behavior of pets through AI-based models.

In recent years, we have observed that the activities of humans [12], [13] and some animals, for example, dogs [14], sheep [15], goats, and horses [16] have been investigated thoroughly; however, there is no research work on cat activity using three kinds of sensors and long short-term memory (LSTM)-based deep learning technology. Therefore, this article explores the activities of cats using the latest and state-of-the-art technologies. Unlike other pet animals, for example, dogs and horses, cats behave in a very complex manner and is a challenging task to observe their behavior. Cats are less cooperative and do not obey the owner as dogs do. They act differently as they do not adhere to a specific activity. Therefore, cats' activity recognition and prediction are very complex, challenging, and, in some cases, impossible. Therefore, to get optimal performance and accuracy, we used three types of sensors, namely, accelerometer, gyroscope, and magnetometer. These sensors could sense the fine details of data related to any particular activity. Most researchers used only one or two sensors for the activity detection of pets, especially for cats [17]. Wernimont et al. [18] used the accelerometer to monitor the behavioral changes in dogs. The study was carried out on six dogs, and they acquired data from them. They applied proprietary behavior recognition algorithms. Yonezawa et al. [19] used a sensing device that has an accelerometer for the detection of the activities of cats. They used Bluetooth to transmit the data and they developed a software for activity detection of cats. They used Bluetooth to transmit the data and they developed a software for activity detection of cats. Ladha et al. [1] developed a platform based on an accelerometer using a statistical classification framework for recognizing dog activities. The 18 dogs participated in this study, and they analyzed the naturalistic behavior of dogs. Their framework achieved 70% classification accuracy. We have used three sensors and applied a deep learning approach for the detection of cats' activity.

The activity detection of cats will help us understanding the behavior of cats and thus enable us to predict the well-being of cats and their overall health condition. It will also provide proactive healthcare recommendations and warnings which means preventing cats from any unhealthy condition or diseases in advance. This article can be summarized as follows. First, we collected the data from ten different cats of different breeds, sizes, genders, and ages. Second, we preprocessed the data which includes labeling of data, removing unnecessary data, data transformation, feature engineering, and data sampling. Third, we developed an LSTM-based deep learning model and trained the model with 80% of the data. We performed hyperparameter tuning of the model to get the optimal performance. Finally, the model was tested with 20% of the testing data. We evaluated the model with performance measures and calculated the accuracy, precision, recall, confusion metrics, and area under curve (AUC) of the model. This model was developed to detect the ten important activities of cats. Those activities are resting, grooming, walking, eating/drinking, vomiting, scratching, running, urinating/pooping, coughing, and jumping. These are the important activities through which the normal routine of the cats can be observed. If there is any unusual change in their activity or behavior, it means, there is something wrong with the cat. In this article, an LSTM-based approach has been adopted to predict the activities of cats. Consequently, these activity trends of cats will provide information about the overall health condition and well-being of cats.

The contributions of this article are as follows.

1) An LSTM-based real-time cat activity detection system has been developed using three kinds of sensor data, i.e., accelerometer, gyroscope, and magnetometer.
2) We are the pioneers who worked on cat activity detection based on deep learning using three sensors' data for the welfare and well-being of cats.
3) Feature engineering has been performed for getting the most important features from the raw data.
4) A long short-term memory (LSTM)-based deep learning model was developed for the detection of ten activities of cats.
5) Hyperparameter tuning was conducted to get the optimal performance of the model.
6) The developed model was tested and deployed in a real-time environment successfully.
7) The model detects the activities of the cat with high accuracy in a real-time environment.

The remaining sections of this article are as follows. Section II describes the related work. Section III deals with the materials and methods we used in this research. Section V explains the experimental results and discussion. Finally, Section VI provides the conclusion.

II. Related Work

Wearable devices have been used for activity detection of humans and animals [7], [20], [21], [22], [23]. Accelerometer, gyroscope, ECG, and magnetometer sensors are incorporated into these wearable devices for the monitoring and detection of activities of animals. Most researchers applied different AI-based algorithms using gyroscopes and accelerometer sensor devices for activity detection.

Kamminga et al. [24] applied deep neural networks for the online activity recognition of goats and sheep. They have used seven different kinds of machine learning classifiers like support vector machine, decision tree, Naïve Bayes, neural networks, linear discriminant analysis, k-nearest neighbors, and deep neural networks. Among these, the deep neural network outperformed other classifiers and obtained an accuracy of 94%.

Chambers et al. [25] developed a deep learning classification algorithm for the detection of activities of dogs using accelerometer data. They have included 2500 dogs in their study and gathered data from them in clinical and real-world settings. They analyzed different activities of dogs, for example, eating, drinking, licking, petting, rubbing, scratching, sniffing, walking, sitting, standing, and shaking. The algorithm

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showed 95.3% true positive for eating and 94.9% for drinking activity.

Hussain et al. [26] developed an LSTM-based model to recognize the ten different dogs' activities, such as walking, down, sitting, staying, sideway, eating, shaking, jumping, nose work, and running. Two different sensors (accelerometer and gyroscope) were used to collect data from numerous breeds of dogs. During preprocessing, a butter low-pass filter was used to reduce the noise and extract crucial features, such as standard deviation, mean, median, and energy measure. To solve the imbalance problem, they trained the LSTM model using preprocessed data and used the class weight method. The proposed model had a testing accuracy of 94.25% and had good accuracy for ten activities.

Kleanthous et al. [22] proposed a deep learning-based intelligent system for the classification of the activities of sheep. Two types of sensors were used for the data collection. They used accelerometer data for extracting the three activities, i.e., active, inactive, and grazing. The model obtained 98.55% and 96.59% accuracy for source data and target data, respectively.

Gutierrez-Galan et al. [27] used a multilayer perceptron-based model for the classification of different behaviors of the horse. Three different gaits of horses were investigated using this model and it achieved an accuracy of 81% in the real scenario. This model is configurable and could be easily embedded into the collar of the animal. Furthermore, it has a longer battery life.

Yen et al. [28] created a deep learning model for the recognition of the daily activities of humans using wearable inertial sensors. Unlike existing wearable devices, i.e., smart watches or wristbands, they used waist wearable devices for data collection. They included 21 participants in their experiment and gathered the data from them who performed six kinds of different activities, namely, sitting, standing, walking, walking downstairs, walking upstairs, and lying. The model reported an accuracy of 97.19% and 93.77% for training and testing, respectively.

Hussain et al. [29] developed a 1-D convolutional neural network (CNN)-based model for the detection of dog activity using sensor data. Their model could classify ten different activities of dogs. The data were collected from ten dogs of different breeds, ages, sizes, and gender. They preprocessed the data before it was used for the training of the model. They used 80% of the data for training and 20% for the testing of the model. The model achieved a training accuracy of 99.70% and a validation accuracy of 96.85%.

Kim and Moon [30] proposed a multimodal data-based method to detect dogs behavior recognition by fusing video data and sensor data (accelerometer and gyrometer). In the video data, they used object detection techniques, such as YOLOv3, YOLOv4, and faster RCNN to identify the movement area of the dogs. Accelerometer and gyroscope data were collected from wearable devices, and various statistical features were extracted from sensor data, such as variance, mean, and standard deviation. The accuracy of multimodal data-based models was higher than the accuracy of single-data-based models on average, with an average of 90.3%. This study can be applied to dog treatment or health monitoring and to improve dog behavior recognition accuracy, an optimal fusion model will be developed in the future.

Colpoys and DeCock [31], in this article, FitBark 2 is used to track dogs' physical activities. This experiment involved 26 dogs wearing the FitBark collar, which recorded their activity in three phases: off-leash room exploration, human–dog interaction, and on-leash walking. A video analysis counted the dogs' steps and compared them to FitBark activity data. FitBark activity and step count showed moderate correlation across all phases, with a high correlation in phase-1and phase-2 and a low correlation in phase-3. According to the study, FitBark can track off-leash dogs' physical activity successfully. However, further research is necessary to improve tracking of on-leash dogs' activity.

Researchers have also investigated the classification accuracy of pet activities by placing wearable devices on different body parts. Some place the sensors on the collar, while others place them on the tail or back of the pets. Kumpulainen et al. [32] examined the behavior classification of dogs by placing the sensor on the collar and harness. They placed two sensors one on the neck collar and the other on the back of the dog. They extracted 54 features from the accelerometer and gyroscope. They compared the performance of four classifiers. They observed that the classification accuracy of the sensor placed on the back of the dog obtained an accuracy of up to 91%. On the other hand, the classification accuracy of the sensor which was placed on the neck yielded 75%.

den Uijl et al. [33] explored eight behavioral states in dogs by mounting the accelerometer sensor on the collar of dogs. They collected data from 51 healthy dogs. The experimental results revealed that the classification accuracy of eating, drinking, trotting, handshake, canter/galloping, and walking could reach up to 95%.

Chapa et al. [34] developed an accelerometer-based system that can detect indicators of animal health and welfare, such as movement, feeding, and resting. In cattle, the accuracy was greater than 95%, while in pigs, the sensitivity was greater than 80%. Currently, there are no commercial accelerometer systems available for pigs. However, accelerometers were used to monitor diseases, oestrus, and calving/farrowing in both pigs and cattle.

Ladha et al. [1] observed the behavior traits of dogs through a set of activities, such as chewing, pawing, and barking. These activities help analyze the well-being of dogs. They collected the data by using an accelerometer sensor mounted on the collar of dogs. They included 18 dogs of different age groups, breeds, and sizes. The classification accuracy of the principal component analysis (PCA)-based feature extraction classifier was around 70%.

Wutke et al. [35] investigate the activities of Pig using video data. They applied a semisupervised neural network for this purpose. For the collection of data, six cameras were used, and the frame rate was 15 frames/s with a pixel dimension of 128×800. They applied two-stage approaches, i.e., training the CNN model for the detection of anomalies and second training the model for the categorization of anomalies. Their model obtained an accuracy of 91%.

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TABLE I
AGE, GENDER, AND BREED OF THE CAT

S.no	Age	Gender	Breed
1	7	Male	Abyssinian
2	5	Female	Maine Coon
3	5	Female	California Spangled
4	7	Male	Chausie
5	6	Female	Cornish Rex
6	3	Female	Chausie
7	5	Male	Maine Coon
8	4	Male	Maine Coon
9	2	Female	Siamese
10	7	Female	Toyger

Griffies et al. [36] used wearable sensor devices for the quantification of two important pruritic behaviors in dogs.

They gathered sensor data and video data from 361 pruritic and normal dogs. The video data were annotated manually by the two persons independently. The data which have the same annotation done by the two observers were included for further analysis. They developed an algorithm for the detection and classification of head shaking and scratching behaviors of dogs. The model could achieve an overall accuracy of 99.24% and 99.56% for scratching and head shaking, respectively.

III. MATERIALS AND METHODS

This section describes the data sources, cat information, feeding, and husbandry environment information, sensors, data collection procedure, and other methodologies used in this article.

A. Data Source

For the experimental purpose, we used Ujura Company cats' data. The data were collected from ten cats for one month. The dataset contained 1 765 825 samples of triaxial data from three sensors, namely, the accelerometer, gyroscope, and magnetometer. The data were collected from 1st November 2021 to 30th November 2021. Thus, this article has analyzed and preprocessed the experimental data in line with the prediction and detection of cats' activities.

B. Cats

A total of ten cats were included in this research study. Among those, four were males, and six were females with different breeds, ages, and sizes. The cats were healthy and in good condition during the experimental period. The detailed information is shown in Table I.

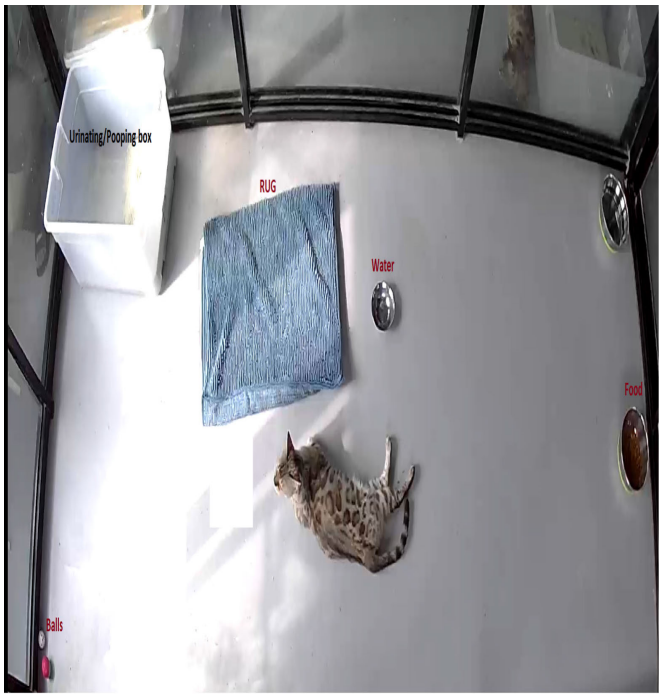

Fig. 1. Feeding and husbandry environment of cats.

C. Feeding and Husbandry

The cats were given a proper diet according to the recommendations from the husbandry and veterinary experts that contained all essential ingredients for their good health. Fig. 1 shows the cats' environment. A very fresh, clean, and spacious environment with proper lighting was provided to all the cats. The experimental rooms were 5.0 × 4.5 m in size. To provide a better environment and to keep them active, the cats were provided with toys and balls to play with. For the scratching of cats, proper rugs were placed in each room. For urinating and pooping a separate box was kept likewise, for rest and sleep a comfortable place was made in each room so that the cats can get an uninterruptable and comfy sleep.

D. Sensor Devices

Wearable sensor devices were used and placed on the neck of each cat using straps for the detection of the activities. The wearable devices consisted of three sensors, namely, accelerometer, gyroscope, and magnetometer. These sensors were able to measure the linear motions, rotational motions, and magnetic motions in all three axes, i.e., x-, y-, and z-axis. The Gyrosensor has a range of ±2000 DPS and a sampling rate of 0.001–100 Hz, and the accelerometer has a range of ±16 g and a sampling rate of 0.001–100 Hz. Likewise, the magnetometer has a range of ±1300 μT (x- and y-axis), and ±2500 μT (z-axis) with a sampling rate of 0.001–25 Hz. The devices weighed 0.2 oz, and the dimensions were 27 × 27 × 4 mm. The sensor devices are designed to detect cat movements of any type like rotational, linear, or magnetic movements. The data are timestamped for synchronization purposes. It has a Lipo battery with 70–100 mAH and a charging time of 2 h. The sensor device is shown in Fig. 2.

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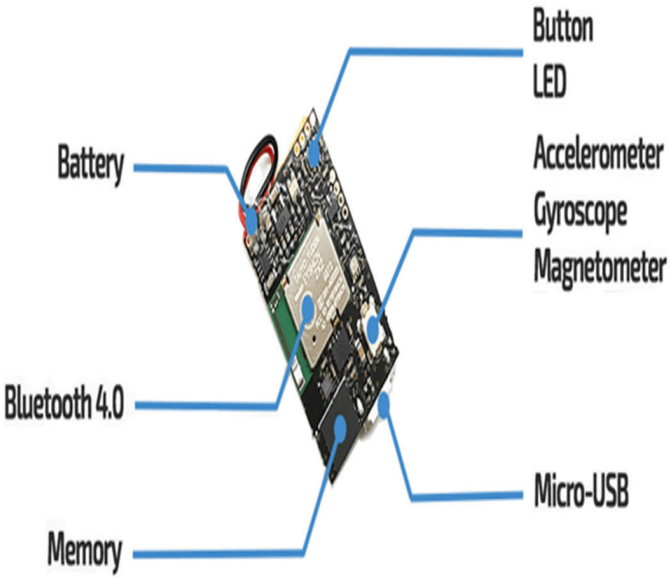
(a)



(b)

Fig. 2. (a) Wearable sensor device. (b) Internal components of the wearable device.

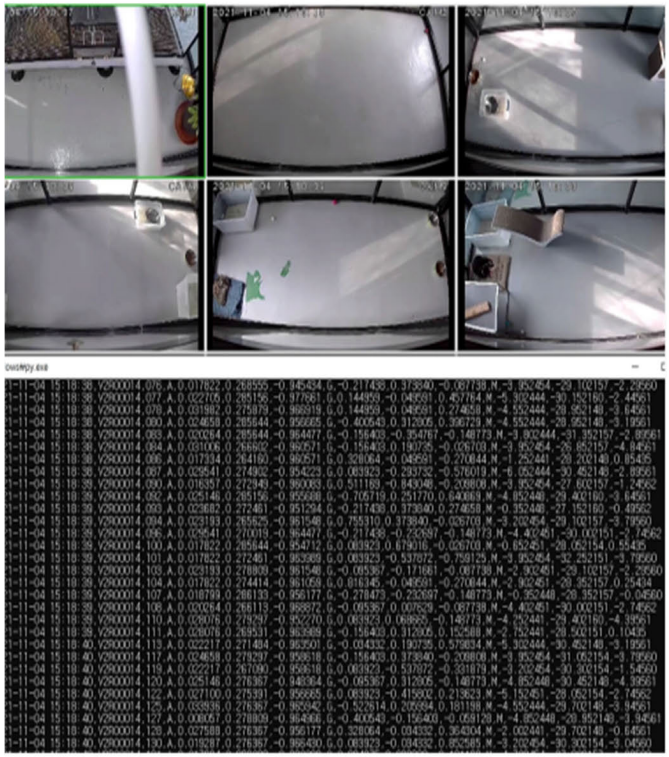
E. Video Recording

The activities of cats were also recorded through cameras as well. The collected data through sensor devices were synchronized with the video for the ground truth and to make sure for the correct labeling of different activities.

F. Data Collection

The data were collected from the cats by placing the sensor devices on the neck of the cats. The data were generated with the movement of the cat. Three kinds of data have been generated, i.e., linear motion data from the accelerometer, rotational motion data from the gyroscope, and magnetic motion data from magnetometer sensors. The data were sent to the server via Bluetooth and were stored on the server, as shown in Fig. 3. At the same time, video recording was also performed for the corresponding sensor data. Moreover, a 25-Hz sampling frequency was used to investigate the activity of cats. The sensor data were labeled while using the video recording.

(a)



(b)

Fig. 3. Data collection procedure. (a) Data center where data is gathered. (b) Cats' environment and data gathering procedure.

The data were collected for ten basic activities of cats: resting, grooming, walking, eating/drinking, vomiting, scratching, running, urinating/pooping, coughing, and jumping.

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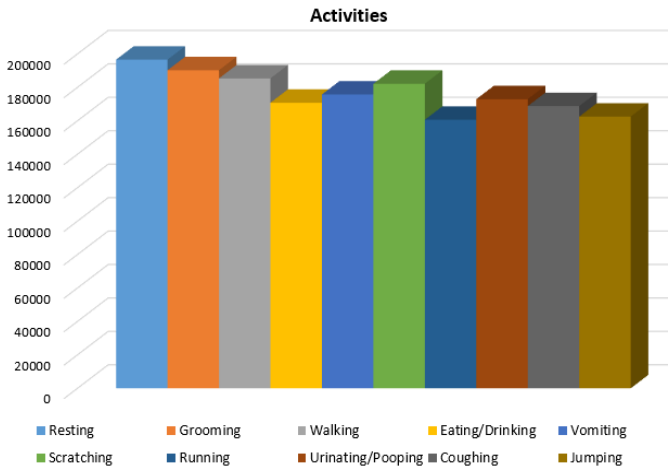

Fig. 4. Data distribution of all the ten activities.

Fig. 4 shows that the data distribution across all ten classes is imbalanced so this behavior of data may render the model overfitting. This issue must be addressed before being used for the model's training. Fig. 5 shows the samples of walking biosignals for all the three sensors.

G. Data Preprocessing

Data preprocessing is one of the important steps in developing an AI-based model [37]. The raw data cannot be used directly for the training of machine learning or deep learning model. We need to refine and cleanse the data. First, we labeled the raw sensor data while watching the videos of cats performing different activities. Once the data were labeled, we need to preprocess the data and remove the noise and anomaly from the data. The data were made fit to be used for the AI model.

H. Data Transformation and Feature Engineering

In order to get important and useful information from the sensor data, we need to apply feature engineering to the sensor data [38]. Feature engineering enables us to extract the most relevant and important features from the data pool. We generated several features from the accelerometer, gyroscope, and magnetometer sensor data. The features were derived while considering a certain number of samples. Some of the important statistical features are standard deviation, mean absolute deviation, mean, minimum, maximum, interquartile range, energy measure, skewness, and kurtosis. Table II shows different stages used for the extraction of important features. AI-based algorithms do not give a good performance, which are trained on raw time series data; therefore, we need to preprocess the raw data for better performance. We have nine basic features of our raw data generated from the three sensors, i.e., accelerometer, gyroscope, and magnetometer. We used some transformation techniques to preprocess the raw time-series data. We applied the windowing technique [39] to transform the data by dividing the data into frame size of five data samples and overlapping two data samples of the previous window, as mentioned in Fig. 6. This windowing helps us to create new features while transforming the data. The newly transformed data are labeled by the activity, which occurs most frequently in the sliding window. Moreover, we also applied the fast Fourier transform technique to transform time-domain data signals into frequency because it gives a better understanding of the sensor data and provides useful information. By applying feature engineering techniques, we generated 312 features. Fig. 6 shows the data transformation procedure, which creates new feature variables from the raw data.

TABLE II
FEATURE EXTRACTION STAGES

S.no	Stage 1	Stage 2	Stage 3
1	Mean	FFT Mean	Index of max value in time domain
2	Standard deviation	FFT Standard deviation	Index of min value in time domain
3	Average absolute deviation	FFT Average absolute deviation	Absolute difference between above indices
4	Minimum value	FFT Minimum value	Index of max value in frequency domain
5	Maximum value	FFT Maximum value	Index of min value in frequency domain
6	difference of maximum and minimum values	FFT Difference of maximum and minimum values	Absolute difference between above indices
7	Median	FFT Median	----
8	Median absolute deviation	FFT Median absolute deviation	----
9	Interquartile range	FFT Interquartile range	----
10	Negative values count	FFT Negative values count	----
11	Positive values count	FFT Positive values count	----
12	Number of values above mean	FFT Number of values above mean	----
13	Skewness	FFT Skewness	----
14	Kurtosis	FFT Kurtosis	----
15	Energy	FFT Energy	----
16	Signal magnitude area	FFT Signal magnitude area	----

I. Data Preprocessing

Sometimes the data samples are not equally distributed, and it is termed imbalanced data. Data sampling is an efficient technique to deal with imbalanced data. The imbalanced data renders the model overfitting or may account for a biased model which does not give the actual accuracy of the model. There are different approaches for balancing the data, for example, random oversampling, synthetic minority oversampling technique (SMOTE), and adaptive synthetic sampling (ADASYN) [40], [41]. However, we used the class weight technique because it helps to increase the cardinality of our dataset and cater for better performance of the model. In this technique, the minority samples are taken more care of while training the model. The loss function is calculated by developing a weighted mechanism. Weights are assigned to the minority and majority classes based on the imbalance scenario in the dataset. A threshold is defined to keep a balance among

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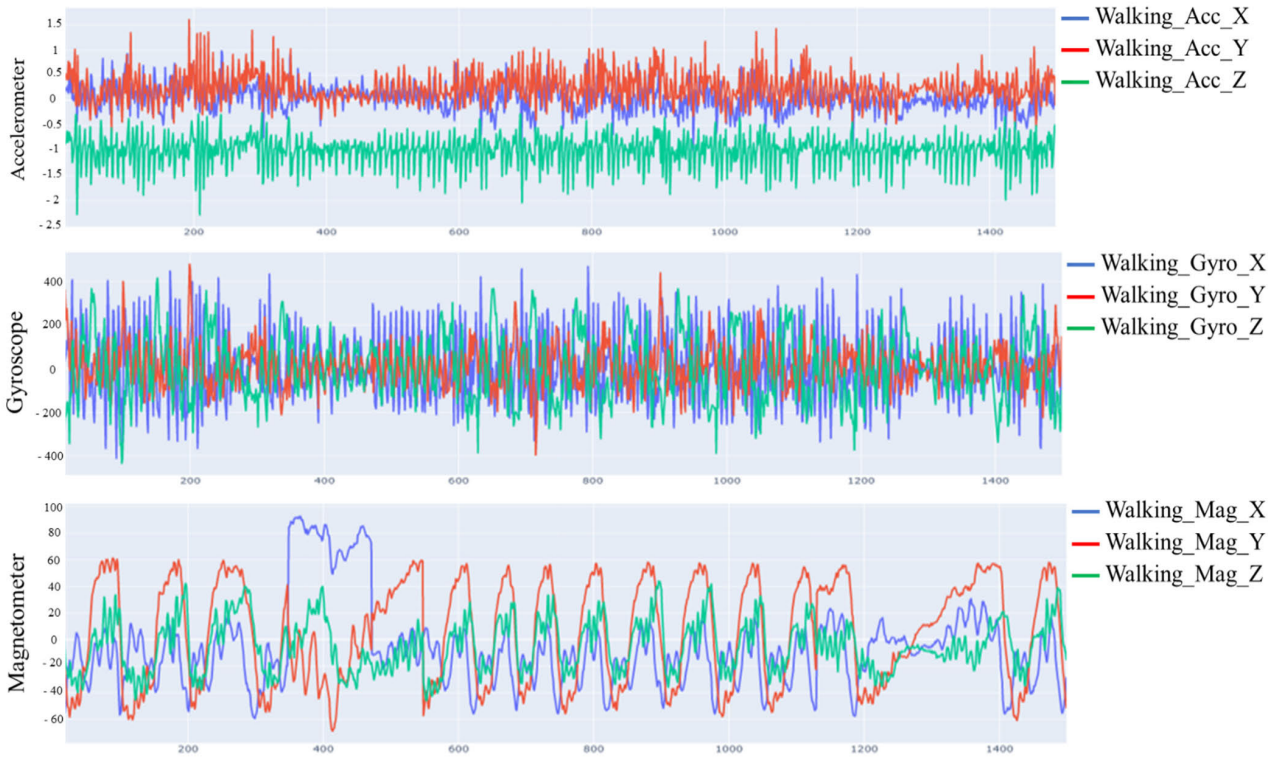

Fig. 5. Samples of biosignals from the wearable device using an accelerometer, gyroscope, and magnetometer for walking activities of cats.

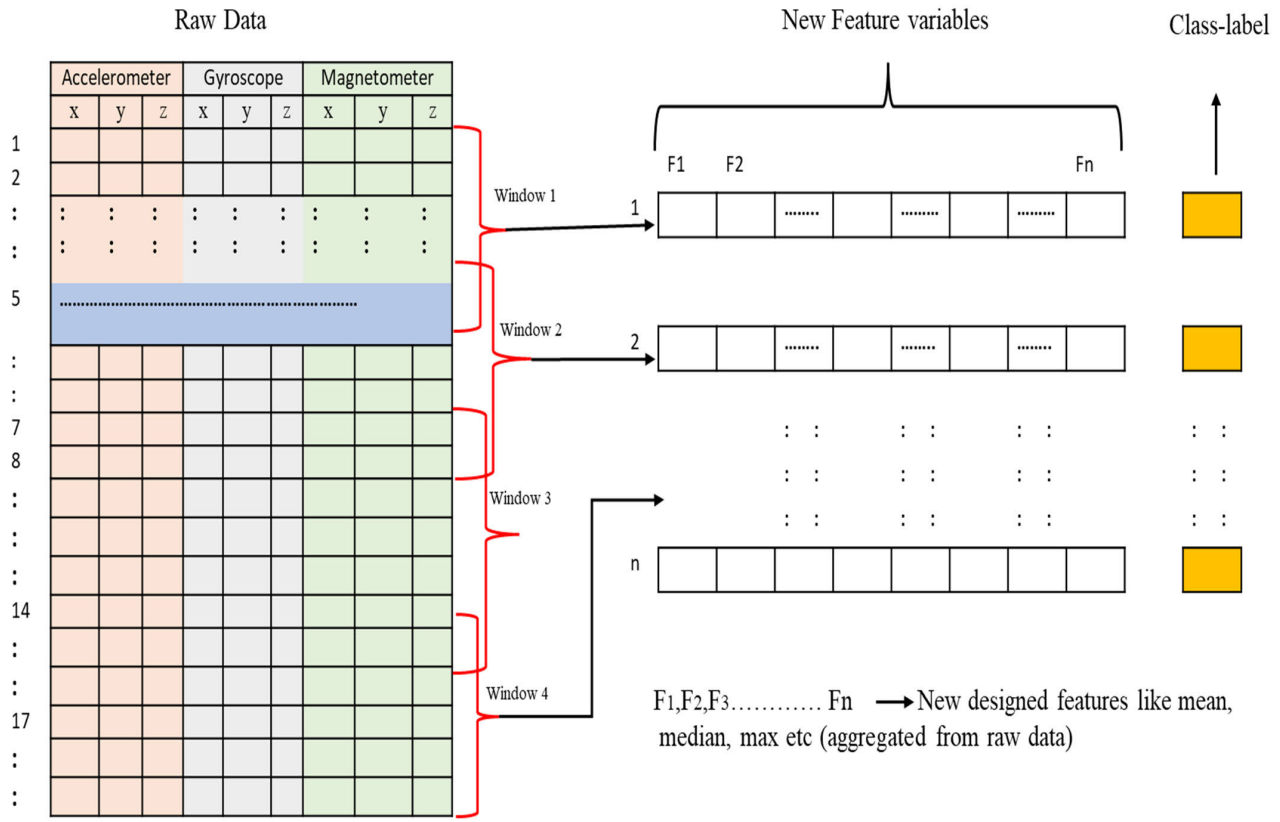

Fig. 6. Data transformation format of the pet activity detection system.

the classes. Table II shows the class weights for each class assigning higher weights to the class with minority samples. For example, running class has fewer samples compared with other classes so it has been assigned higher weights using the

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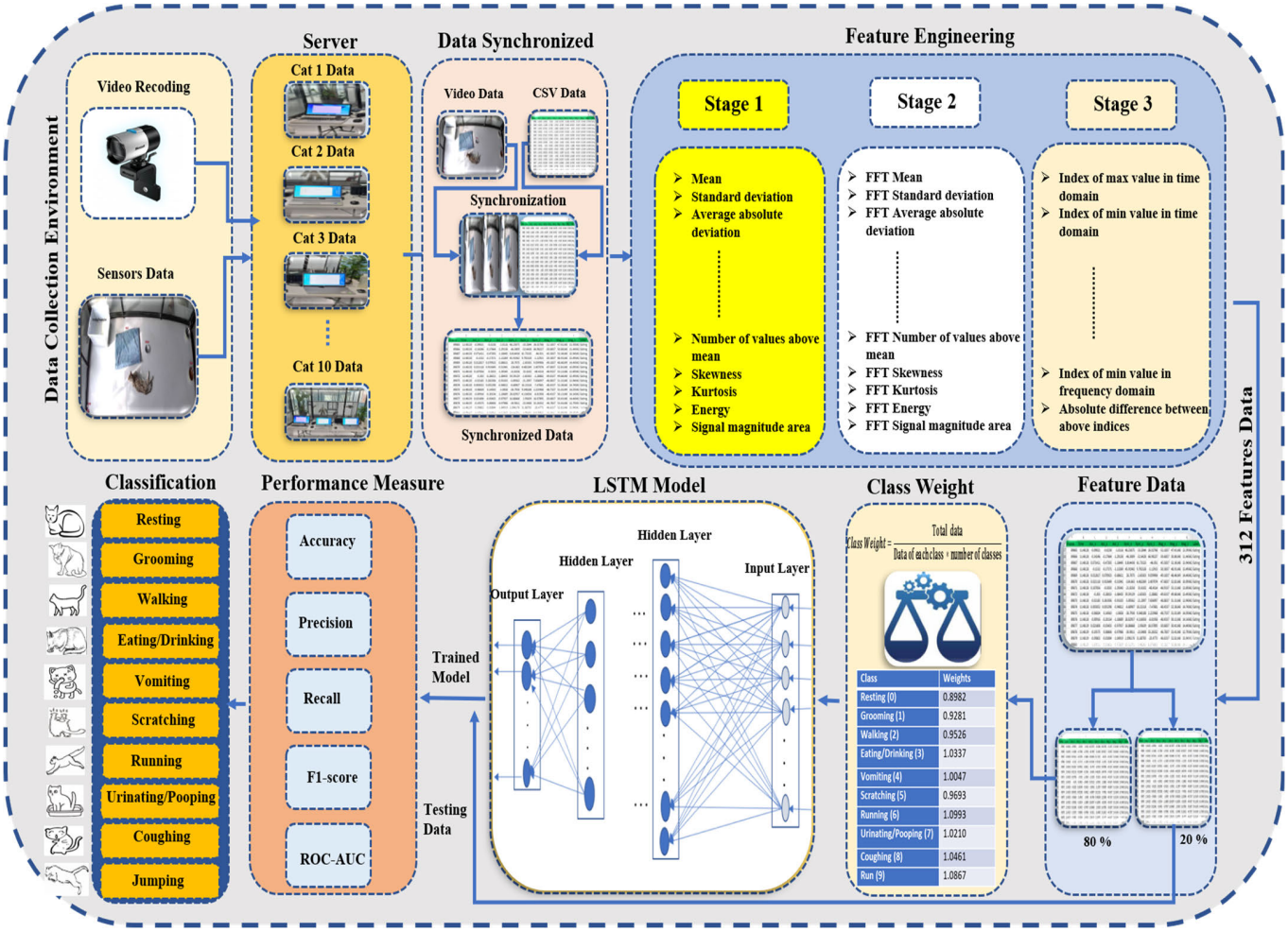

Fig. 7. Architecture of LSTM-based pet activity detection system.

following formula:

$$wi = \frac{n\text{instances}}{(n_\text{classes} \times n_\text{instances}i)} \tag{1}$$

where wi represents the class weight for each class, n_instances represents the total number of data samples, and n_classes shows the number of classes.

IV. Architecture of the Proposed Research Study

The whole architecture of our research study has been discussed below and is shown in Fig. 7. The complete development process can be summarized as follows. First, the activity data were extracted using wearable sensor devices from cats of different ages, sizes, genders, and breeds. Second, the data were labeled for the ten activities, i.e., resting, grooming, walking, eating/drinking, vomiting, scratching, running, urinating/pooping, coughing, and jumping using the video recordings of the same activities to generate the ground truths. Third, preprocessing of data was performed, and the noise and other redundant data were removed from the dataset. In order to remove the noise from the data butter low-pass filter was used. Fourth, to get the insight knowledge and the hidden patterns of data, featuring engineering was conducted, and as a result, multiple features were generated. The dataset was divided into the training data of 80% and the testing data of 20% after preprocessing the dataset. Fifth, an LSTM-based model was developed and trained. Then, hyperparameter tuning of the model was conducted. Sixth, the performance evaluation [accuracy, precision, recall, $F1$-score, and the area under the ROC curves (AUC)] of the proposed model was performed to analyze the validity of our proposed model. Finally, the model predicted all ten activities of cats as its output with optimal performance.

A. Proposed Activity Detection Model

LSTM is a type of deep learning having feedback connections. It has been introduced to address the vanishing gradient issue in recurrent neural networks (RNNs) because RNN cannot keep tracking of the longer sequences of data. It consists of cells and different kinds of gates, i.e., input gate, output gate, and forget gate. The internal mechanism which regulates the flow of information is called gate. These gates analyze the sequence of information and decide whether to keep it or throw it away. The input data are transformed into machine-readable vectors and is passed down from the previous hidden state to the next step in the network. The input layer receives the input features and processes them to

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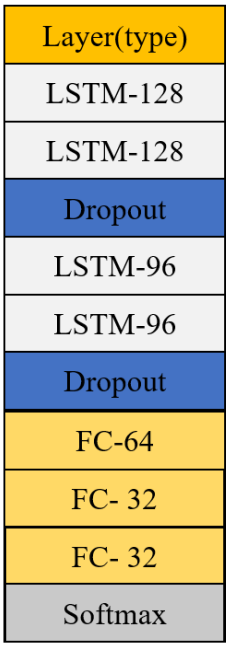

Fig. 8. Internal architecture of LSTM-based pet activity detection model.

the next hidden layers, where computations take place and process them further and deliver them to the output layer. At the output layer, classification results are produced, and we get the final classification of the data. The cells of hidden layers are the computational units that have one or multiple weighted inputs connected with them. A transfer function is used for the combination of all the inputs, and the output of the transfer function is produced at the output layer. The output signal must be capable of predicting the actual class for each of the input features by indicating a higher output value for the actual class than the rest of the classes. The internal architecture of our proposed LSTM-based pet activity detection system is briefly elaborated in Fig. 8.

The proposed LSTM-based pet activity detection system was developed with Four hidden layers and nodes were 128, 128, 96, and 96. The input layer receives the data and propagates them to the hidden layers. To get the optimal hyperparameters, we conducted the hyperparameter tuning. Rectified linear unit (ReLU) was used as an activation function because it gave optimal accuracy. Furthermore, the "Adam" optimizer was used for the weight optimization that is based on stochastic gradient and minimized the network error. "Categorical cross-entropy" was used as the loss function. The input layer received 312 input variables consisting of different features. The input data were propagated to the next layers in the network by the feedforward mechanism of neural networks. The training of the model was improved with the back-propagation mechanism. The output layer was incorporated with the SoftMax activation function for the classification of all the ten activities and finally produced the result.

B. Performance Measure

To evaluate the performance of deep learning algorithms, we used certain metrics, such as accuracy, precision, recall, and $F1$-score, as shown in Table III

TABLE III
CLASS WEIGHTS FOR EACH CLASS

Class	Weights
Resting (0)	0.8982
Grooming (1)	0.9281
Walking (2)	0.9526
Eating/Drinking (3)	1.0337
Vomiting (4)	1.0047
Scratching (5)	0.9693
Running (6)	1.0993
Urinating/Pooping (7)	1.0210
Coughing (8)	1.0461
Jumping (9)	1.0867

$$\text{Accuracy} = \text{TP} + \text{TN}/\text{TP} + \text{FP} + \text{FN} + \text{TN} \tag{2}$$

$$\text{Recall} = \text{TP}/\text{TP} + \text{FN} \tag{3}$$

$$\text{Precision} = \text{TP}/\text{TP} + \text{FP} \tag{4}$$

$$F1\text{-Score} = 2 \times (\text{Recall} \times \text{Precision})/(\text{Recall} + \text{Precision}) \tag{5}$$

where TP, TN, FP, and FN are denoted by the true positive, true negative, false positive, and false negative, respectively.

C. Implementation Environments

All these experiments and analyses were performed using a 64-bit Windows operating system with Intel[1] Core[2] i5-12500 CPU, 64-GB RAM, and NVIDIA GeForce RTX 3080 GPU. Jupiter Notebook, an open-source web application, was used for the implementation and development of the proposed LSTM model by using Python language (Version 3.8.3), Tensor flow, Keras (Version 2.8.0), and Scikit-learn libraries.

V. EXPERIMENTAL RESULTS AND DISCUSSION

The learning curve of the proposed pet activity detection system has been shown in Fig. 9 by representing the relationship between accuracy and loss for the training and validation dataset. It is evident from Fig. 9 that as the number of epochs increases, the loss decreases, and resultantly, the accuracy of the model increases. We use an early stopping parameter; therefore, when the model did not learn more during certain epochs, it stopped training. We use early stopping for five epochs. So if the performance of the model does not increase during five consecutive epochs the model training will stop automatically. Furthermore, in Fig. 9(a), the model's accuracy for training and validation

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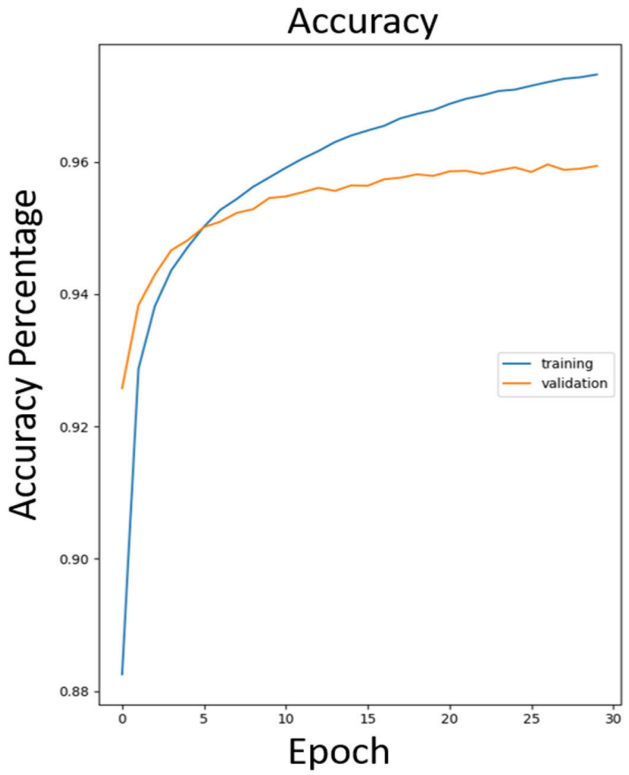
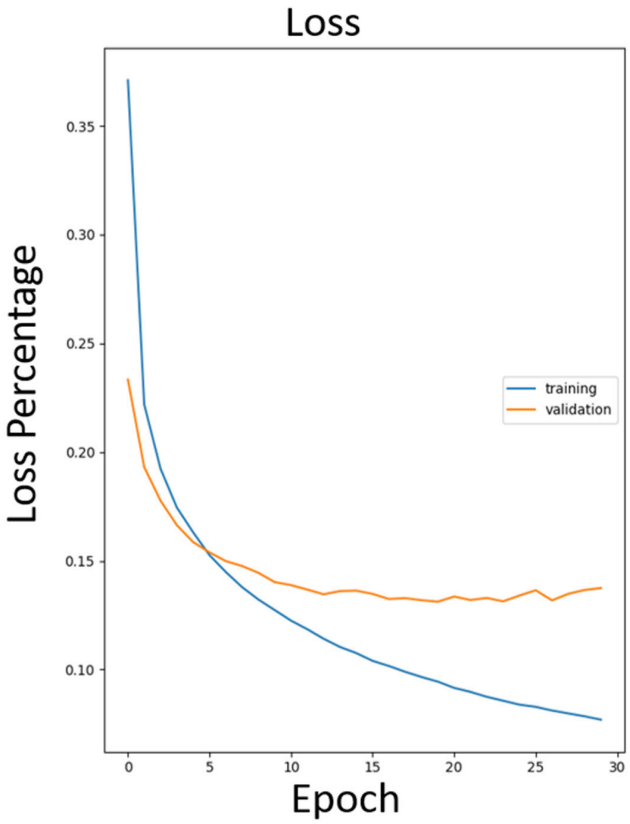


(a)



(b)

Fig. 9. Learning curve of the LSTM model for the activity detection of cats. (a) Learning curve of accuracy. (b) Learning curve of loss.

increases suddenly during the first few epochs, and then, it shows gradual change as the number of epochs increases. The accuracy and validation reached 98.3% and 96.6%, respectively. Likewise, in Fig. 9(b), the loss of the model for training and validation dwindles abruptly during the first few epochs and gets a flattened curve with an increasing number of epochs. So, it can be concluded from the learning curve graph that neither there is an overfitting nor under-fitting problem during the model's training.

TABLE IV
COMPARISON WITH OTHER DEEP LEARNING MODELS

Classifiers	Accuracy	Precision	Recall	F1-score	AUC
Proposed LSTM	**0.9594**	**0.9591**	**0.9588**	**0.9589**	**0.9983**
1D CNN	0.9516	0.9515	0.9511	0.9513	0.9975
ANN	0.9335	0.9325	0.9327	0.9318	0.9959

Note: LSTM denotes Long short-term memory; 1D CNN 1D convolutional neural network; ANN Artificial neural network.

TABLE V
COMPARISON WITH TWO- AND FOUR-LAYER LSTM MODELS

S. no	Sensor	Accuracy		ROC-AUC	
		4-Layer LSTM	2-Layer LSTM	4-Layer LSTM	2-Layer LSTM
1	Acc	83.76	79.27	98.22	97.01
2	Gyro	68.65	62.74	94.74	92.94
3	Mag	88.86	83.20	99.06	97.77
4	Acc, Mag	92.34	88.27	98.21	97.05
5	Gyro, Mag	91.93	87.69	99.42	98.50
6	Acc, Gyro	87.70	83.49	98.90	97.90
7	**Acc , Mag, Gyro**	**95.94**	91.31	**99.83**	99.15

Note: Acc denotes accelerometer, Gyro: gyroscope, Mag: magnetometer

In Table IV, we have shown the accuracy, precision, recall, $F1$-score, and AUC of each model which we applied in our experiment. The experimental results showed that the LSTM-based model outperformed all other models.

We also performed a comparative analysis while combining different sensors and different LSTM architectures. Table V shows the details of the comparative analysis.

We also calculated the confusion matrix for our proposed LSTM system showing its classification result for each class. Fig. 10 shows the confusion matrix of all ten activities of cats. It represents the actual and the predicted classification on the x-axis and y-axis, respectively. The class labels are resting, grooming, walking, eating/drinking, vomiting, scratching, running, urinating/pooping, coughing, and jumping. The values in the diagonal represent the correct classification, while the values other than the diagonal values are misclassifications. There are some false detections for some activities which are similar to each other. For example, sometimes jumping and running both activities generate similar data trends; therefore, there are more false detections between running and jumping as you can see from the confusion matrics. The same goes for walking and running. So model confused between those activities and misclassified the activities.

A receiver operating characteristic (ROC) is a graphical representation of the diagnostic ability of the classifiers. It is used for the comparison of diagnostic tests. A graph is plotted by considering the true positive rate against the false-positive rate. The more the curve is closer to the top-left corner, the higher will be the performance. The ROCs of all ten activities have been shown in Fig. 11. The study we proposed was

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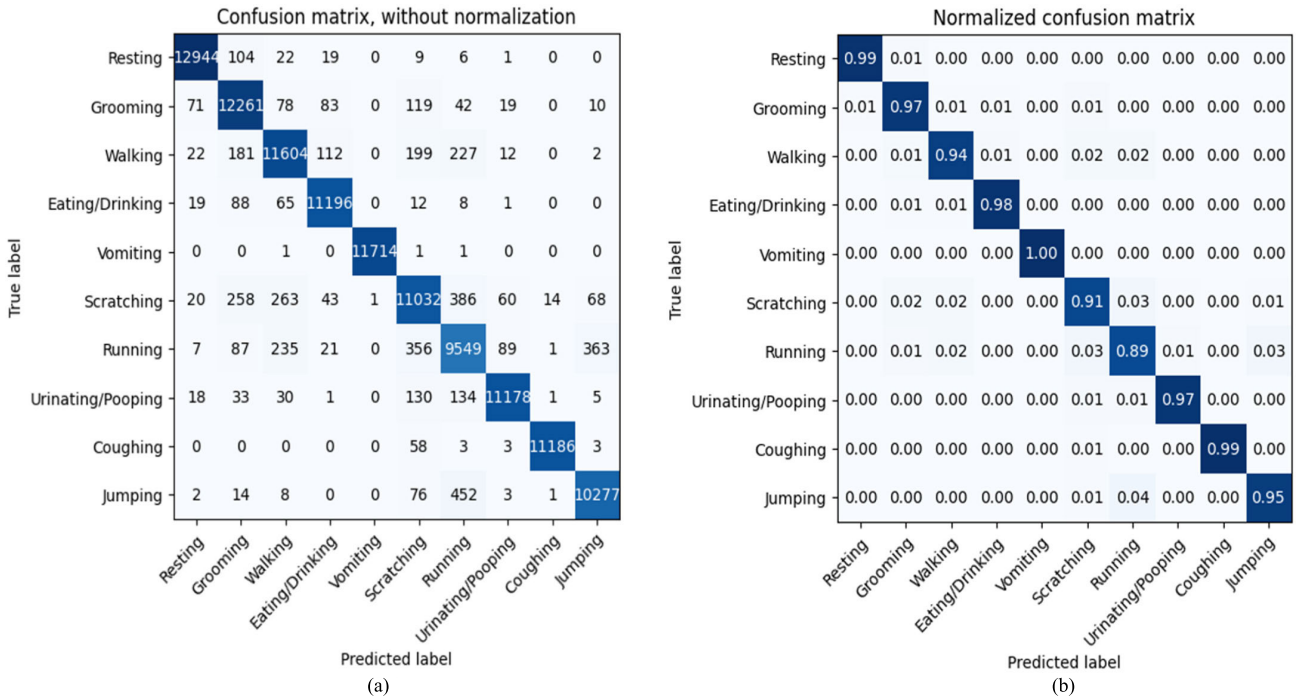

Fig. 10. (a) and (b) Confusion matrix of cats' activity detection without and with normalization, respectively.

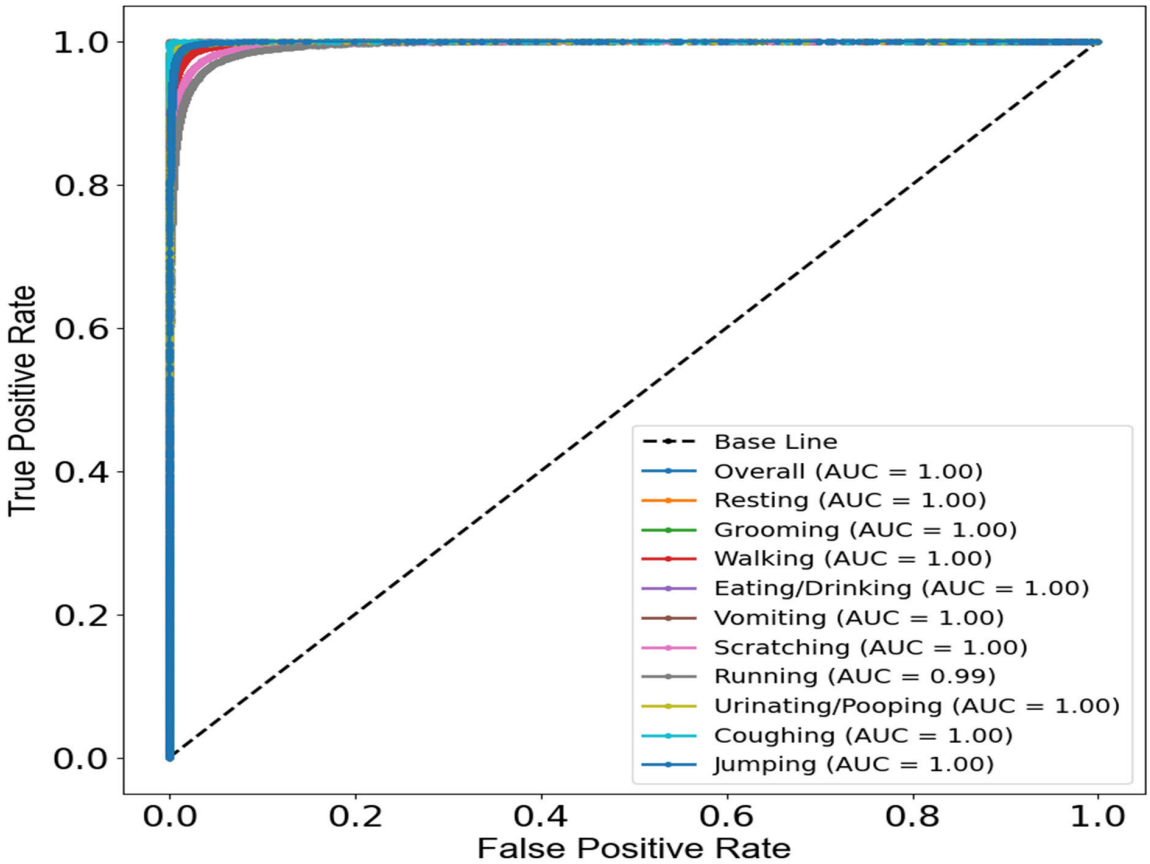

Fig. 11. ROC–AUC curve of all the activities.

compared with a number of other papers published in the past. This study found that the developed system performed better than all previous studies based on cats. Table VI shows a comparative analysis of these proposed studies.

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TABLE VI
COMPARISON OF OUR RESULT WITH OTHER PUBLISHED PAPERS

Reference	No of activity	Animal	Outcomes	Performance indices
[1]	17	Dog	68.6%	Accuracy
[26]	10	Dog	>95%	Accuracy
[33]	8	Dog	94.25%	Accuracy
Our	10	Cat	95.94%	Accuracy

A. Limitations and Future Work

There are certain limitations to this research study; first, we had limited number of cats. We have collected data from ten different cats of different ages, breeds, and genders. Although we have collected a plethora of data from the available cats, however, increasing the number of cats providing a variety of data may enhance the generalizability of the model. In the future, we include more cats of different breeds, ages, and gender to the study and collect data. Second, in this study, we have gathered the data by placing the sensors on the neck. In the future, we place the sensors on the tail and on the waist of cats and analyze the activities of cats. Furthermore, our future work will also focus to investigate the pooping/urinating and eating/drinking activities separately and add more activities.

VI. CONCLUSION

This article proposed an LSTM-based detection system for the prediction and classification of ten basic activities of cats. We collected the data from different cats of different ages, breeds, and genders. The raw data were preprocessed and transformed into the proper format that is to be used for training the model. We also applied feature engineering to get the meaningful and most important features from the raw data. We developed an LSTM-based model for the detection of activities of cats. This detection system will surely help to monitor the activities of cats and will ensure their well-being and overall health. To the best of our knowledge, this article is the first of its kind which is based on an LSTM-based model for cat activity detection using three kinds of sensors data accelerometer, gyroscope, and magnetometer.

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